

Library Circulation System (LCS)

The Library Circulation System is designed to automate and simplify the day-to-day operations of a university library. Traditionally, most library activities such as borrowing, returning, and tracking books were performed manually, which often led to delays, errors, and loss of records. To overcome these limitations, the Library Circulation System provides a computerized solution that ensures efficient book management and better service for both students and librarians.

In this system, every student and faculty member is registered as a library user. Each user is provided with a unique library ID, which allows them to log into the system and access its services. Once logged in, a student can search for books by title, author, or subject. The system instantly displays a list of matching books along with their availability status—whether the book is currently available, issued to someone else, or reserved.

If the desired book is available, the student can request to borrow it. The librarian verifies the student's eligibility (for example, checking if the student has any overdue books or unpaid fines) and then issues the book. The system automatically records the issue date and calculates the due date, typically seven or fourteen days later. The book's status changes from "Available" to "Issued," ensuring that other users know it is currently borrowed.

When the student returns the book, the librarian scans the book's ID, and the system updates its status back to "Available." If the book is returned after the due date, the system calculates the overdue fine based on predefined rules and displays it on the student's account. This automation saves time and ensures accuracy in fine management.

Sometimes a book may not be available because it has already been borrowed by another member. In that case, the student can place a reservation through the system. When the book is returned, the system automatically notifies the student via email or SMS that the book is ready for collection. This reservation feature prevents disputes and ensures fairness among library users.

The librarian also benefits from the system's ability to generate various reports—such as a list of issued books, overdue items, popular books, and member activity summaries. These reports help in managing inventory, understanding user demand, and planning new book purchases.

Overall, the Library Circulation System ensures smooth and transparent operations, reduces manual workload, and enhances user satisfaction by providing a modern, efficient, and reliable library experience.